Advait Pandharpurkar

Forward Deployed AI Engineer · Robotics Perception, Teleoperation & Edge Deployment

Pune, India · Open to relocation · +91 82759 93693 · advait.low@gmail.com · linkedin.com/in/lordofwizard · github.com/lordofwizard

SUMMARY

AI engineer with 2+ years shipping perception, teleoperation and embedded software onto real robots: an NVIDIA Jetson depth perception stack, internet teleoperation of ROS2 robots, and camera streaming from production robot hardware. Strong Python and C++ on Linux across 3 robot platforms, with industrial control and on-site customer debugging.

TECHNICAL SKILLS

Languages: C++, Python, Rust, Bash, Java, JavaScript
Robotics & Perception: ROS2, ROS, OpenCV, depth perception, point clouds, camera calibration, NVIDIA Jetson, edge deployment, teleoperation, real robot platforms (Cepheus, Diadem, TTB-PRO-MAX)
ML & Model Lifecycle: model fine-tuning, training and evaluation pipelines, dataset curation, PyTorch, TensorFlow, YOLO, SAM, vision transformers, DINOv2, CLIP, grasp generation (GraspNet, Dex-Net), VLA models, diffusion policies, imitation learning, TensorRT, ONNX, quantization, MLOps
Systems & Industrial: Linux, PROFINET, industrial controls, Docker, Kubernetes, Git, CI/CD, FastAPI, REST APIs, Microservices, Azure DevOps
Cloud & Data: Google Cloud, Azure, AWS, Databricks, data pipelines, rosbag and teleoperation data capture

EXPERIENCE

Associate Software Platform Engineer II (AI Engineer) — MetLife, Corporate Technology Dec 2025 – Present

- Build a 7-tool MCP AI service for Azure DevOps (pipeline status, changelogs, suggested code fixes) plus the internal MCP server platform other product teams run their agents on.
- Take ambiguous requirements to working proofs of concept, including a Reinsurance data-validation POC that secured funding for 2 AI engineering hires and a Copilot cost and usage plugin now demoed to leadership.
- Train the full MGCC CorpTech organization on prompt engineering across 3 enablement topics, turning hands-on agent experience into reusable guidance for other engineers.

Software Developer Intern — Vanderlande Aug 2024 – Mar 2025

- Removed the dedicated industrial PC from loop-sorter test emulation by writing a C++ and Rust PROFINET translation layer that runs on a VM, reaching 90% sorter efficiency against the real control pipeline.
- Evaluated 4 IO-emulation designs and rejected the full IO-layer redesign as too slow to land, shipping a thinner translation layer that worked with the existing controls stack instead.

Software Developer Intern — RigBetel Labs Dec 2023 – Jul 2024

- Enabled internet teleoperation of ROS2 robots through 1 Google Cloud-hosted Zenoh router, replacing LAN-only control and bridging DDS pub/sub across NAT.
- Benchmarked 4 video transports (WebRTC, TCP, UDP, Zenoh-DDS) and shipped WebRTC camera middleware streaming NDI feeds at 720p/30 fps from real robot hardware.
- Built RoboNet, a Rust and Python robot networking suite (hotspot daemon, logs web server, FastAPI config API) now running on every RigBetel Labs robot across 2 compute platforms.

Freelance Robotics Consultant — Vector Robotics Feb 2025

- Delivered 1 on-device depth perception stack for OAK-D cameras in Rust and OpenCV, optimised to run on NVIDIA Jetson edge hardware and published through a ROS2 Jazzy package, demoed live at IIT Bombay E-Summit.

EDUCATION

Bachelor of Engineering, Computer Engineering — Pune University 2025